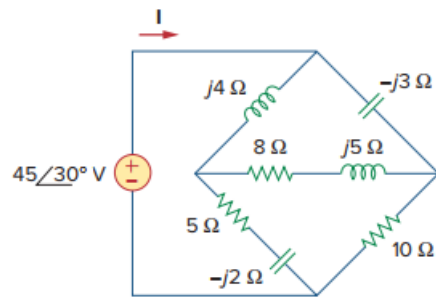


Problem

Find I in the circuit of Fig. 9.30.

Figure 9.30:



Step-by-step solution

Step 1 of 9

Refer to Figure 9.30 in the textbook.

Draw the modified circuit with delta representation.

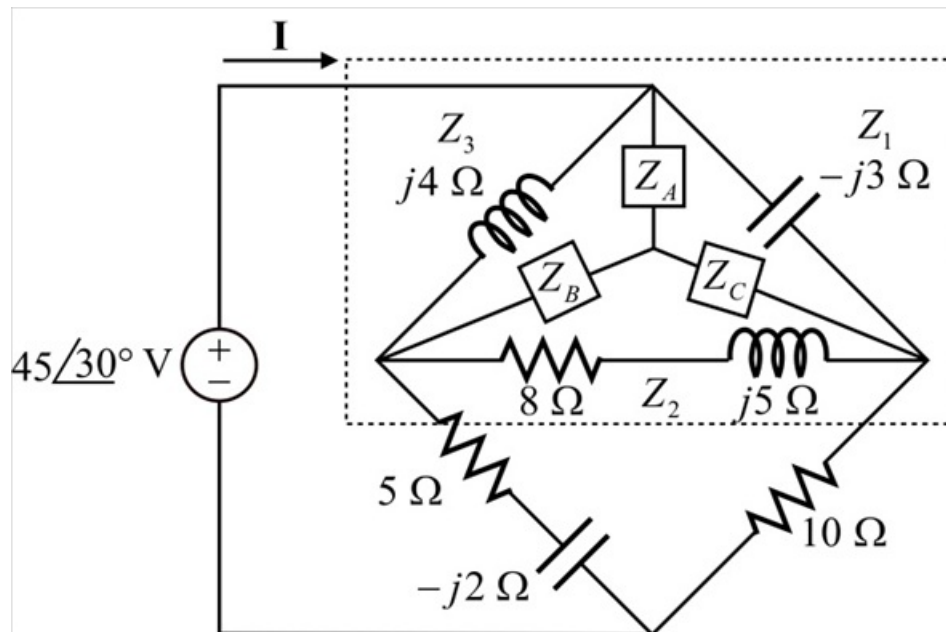


Figure 1

Step 2 of 9

Consider the following star connected impedances.

$$Z_1 = -j3$$

$$Z_2 = 8 + j5$$

$$Z_3 = j4$$

Calculate the value of the impedance, Z_A .

$$Z_A = \frac{Z_1 Z_3}{Z_1 + Z_2 + Z_3}$$

Substitute $-j3$ for Z_1 , $8 + j5$ for Z_2 and $j4$ for Z_3 .

$$\begin{aligned} Z_A &= \frac{(-j3)(j4)}{-j3 + 8 + j5 + j4} \\ &= \frac{12}{8 + j6} \\ &= 0.96 - j0.72 \, \Omega \end{aligned}$$

Step 3 of 9

Calculate the value of the impedance, Z_B .

$$Z_B = \frac{Z_2 Z_3}{Z_1 + Z_2 + Z_3}$$

Substitute $-j3$ for Z_1 , $8 + j5$ for Z_2 and $j4$ for Z_3 .

$$\begin{aligned} Z_B &= \frac{(8 + j5)(j4)}{-j3 + 8 + j5 + j4} \\ &= \frac{-20 + j32}{8 + j6} \\ &= 0.32 + j3.76 \, \Omega \end{aligned}$$

Step 4 of 9

Calculate the value of the impedance, Z_C .

$$Z_C = \frac{Z_1 Z_2}{Z_1 + Z_2 + Z_3}$$

Substitute $-j3$ for Z_1 , $8 + j5$ for Z_2 and $j4$ for Z_3 .

$$\begin{aligned} Z_C &= \frac{(-j3)(8 + j5)}{-j3 + 8 + j5 + j4} \\ &= \frac{15 - j24}{8 + j6} \\ &= -0.24 - j2.82 \, \Omega \end{aligned}$$

Step 5 of 9

Draw the following modified circuit:

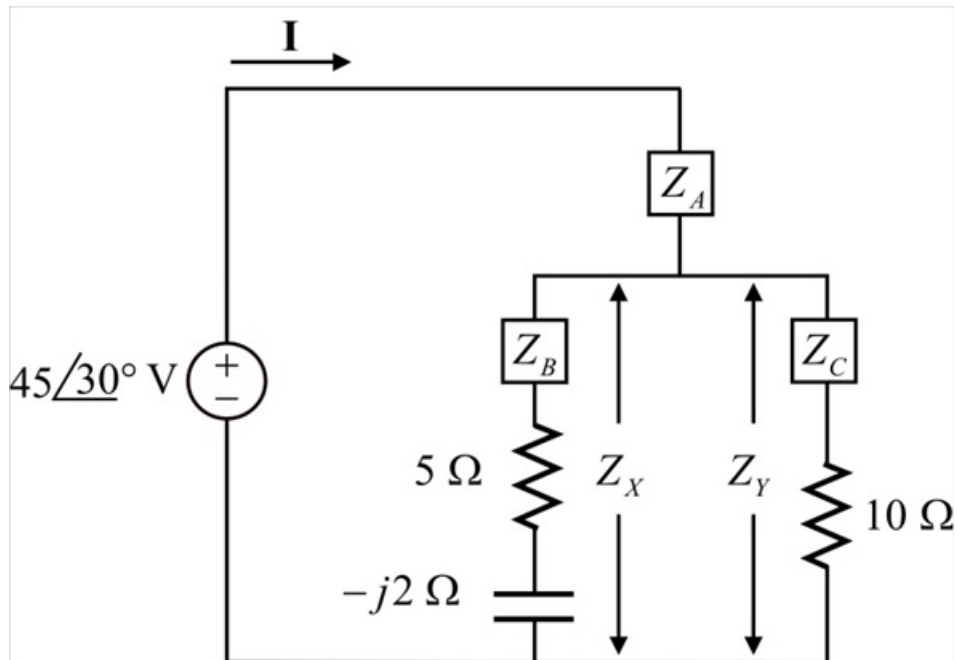


Figure 2

Step 6 of 9

Calculate the value of the impedance, Z_X .

$$Z_X = Z_B + 5 - j2$$

Substitute $0.32 + j3.76 \Omega$ for Z_B .

$$\begin{aligned} Z_X &= 0.32 + j3.76 + 5 - j2 \\ &= 5.32 + j1.76 \end{aligned}$$

Calculate the value of the impedance, Z_Y .

$$Z_Y = Z_C + 10$$

Substitute $-0.24 - j2.82 \Omega$ for Z_C .

$$\begin{aligned} Z_Y &= -0.24 - j2.82 + 10 \\ &= 9.76 - j2.82 \end{aligned}$$

Step 7 of 9

Here Z_X and Z_Y are in parallel.

Calculate the value of the impedance, Z_Z .

$$Z_Z = Z_X \parallel Z_Y$$

Substitute $5.32 + j1.76$ for Z_X and $9.76 - j2.82$ for Z_Y .

$$\begin{aligned} Z_Z &= (5.32 + j1.76) \parallel (9.76 - j2.82) \\ &= \frac{(5.32 + j1.76)(9.76 - j2.82)}{(5.32 + j1.76) + (9.76 - j2.82)} \\ &= \frac{56.8864 + j2.1752}{15.08 - j1.06} \\ &= 3.7437 + j0.4074 \end{aligned}$$

Step 8 of 9

Draw the following modified circuit:

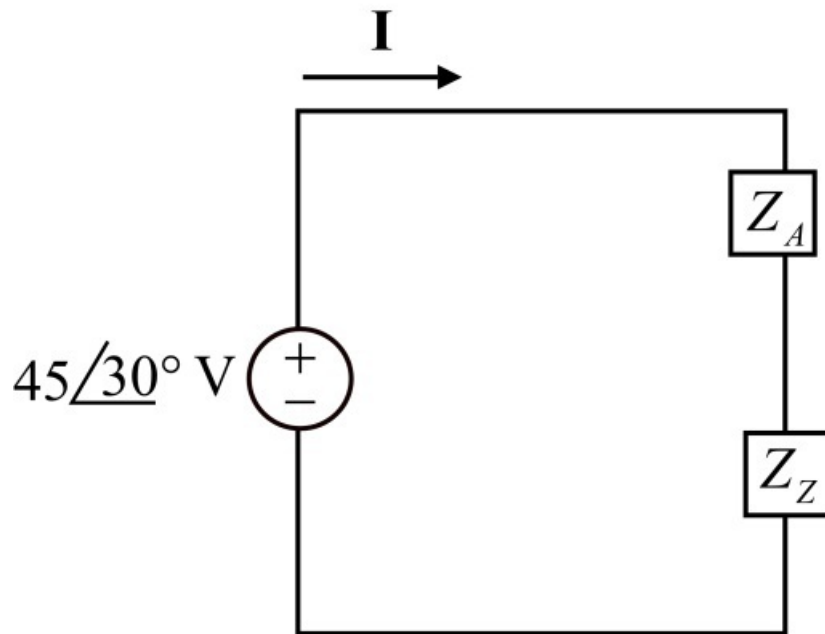


Figure 3

Step 9 of 9

Calculate the circuit current.

$$\mathbf{I} = \frac{45\angle 30^\circ}{Z_A + Z_Z}$$

Substitute $0.96 - j0.72 \, \Omega$ for Z_A and $3.7437 + j0.4074$ for Z_Z .

$$\begin{aligned}\mathbf{I} &= \frac{45\angle 30^\circ}{Z_A + Z_Z} \\ &= \frac{45\angle 30^\circ}{0.96 - j0.72 + 3.7437 + j0.4074} \\ &= \frac{45\angle 30^\circ}{4.7037 - j0.3126} \\ &= \frac{45\angle 30^\circ}{4.714\angle -3.8^\circ}\end{aligned}$$

$$\mathbf{I} = 9.546\angle 30.8^\circ$$

Therefore, the value of the current, $\mathbf{I}$ is $9.546\angle 30.8^\circ$.